

DC Motor Characterization and Inductance Estimation

ESP32-Based Current Sensing, Equivalent Circuit Modeling, and Exponential Transient Analysis

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Technical Documentation Report

Executive Summary

This project characterized the transient electrical behavior of a small 3 V DC motor using an ESP32-controlled switching circuit, shunt-based current sensing, and exponential curve fitting. The motor was modeled as a first-order electromechanical load using winding resistance, inductance, and back electromotive force (back-EMF).

The collected ADC data was fit to a first-order exponential response. The fitted time constant was 20.3 μ s. Using the measured motor resistance of 3.9 Ω , the estimated motor inductance was 79.2 μ H. The fitted curve matched the measured transient shape with $R^2 = 0.939$ and RMSE = 45.0 ADC counts.

1. Project Overview

The goal of this project was to build a repeatable method for estimating the inductance of a small DC motor. A motor winding behaves electrically like a resistance in series with an inductance, while the rotating motor generates an opposing back-EMF. By switching the motor on and measuring the current-related ADC response over time, the winding inductance can be estimated from the transient response.

The project combined circuit modeling, breadboard prototyping, ESP32-based sampling, MATLAB curve fitting, and result validation against a simplified LTspice-equivalent model.

2. Objectives

- Develop a simplified electrical model for a 3 V DC motor using resistance, inductance, and back-EMF.
- Build a low-voltage test circuit that switches the motor and senses current using a shunt resistor and op-amp stage.
- Use an ESP32 to sample the current-sense signal at controlled delay intervals during motor turn-on.
- Fit the measured ADC response to a first-order exponential curve and extract the time constant.
- Estimate the motor inductance using the relationship between resistance, inductance, and time constant.
- Document the circuit, test approach, data-processing method, results, limitations, and lessons learned.

3. Theory and Equivalent Circuit Model

At steady state, the voltage across a DC motor can be approximated as the sum of the back-EMF and the resistive voltage drop across the winding resistance:

$$V_m = e_{back} + iR$$

This steady-state model is incomplete during turn-on because the motor winding has inductance. The transient model includes the inductive voltage term:

$$V_m = e_{back} + iR + L(di/dt)$$

The back-EMF term is related to angular velocity. For a motor, it can be represented as:

$$e_{back} = K_v * \omega$$

For the test model, the back-EMF was treated as a measured constant of 1.42 V during simulation. The measured winding or load resistance used for the inductance calculation was 3.9 ohm.

For a first-order RL step response, the time constant is related to resistance and inductance by:

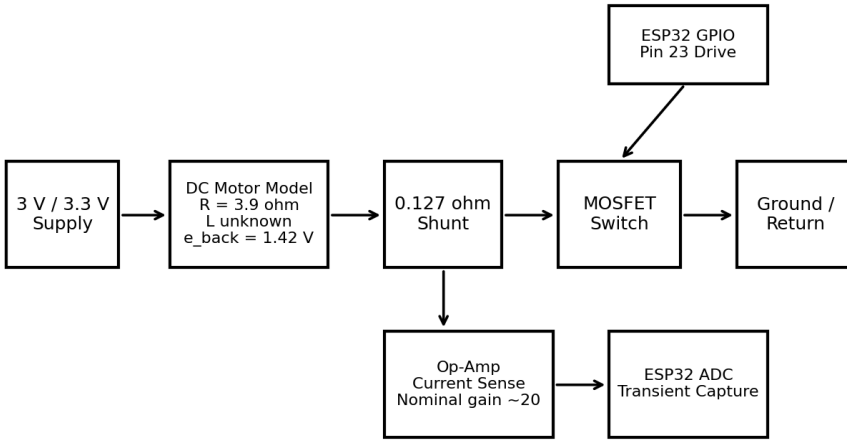
$$\tau = L / R \quad \text{therefore} \quad L = R * \tau$$

This relationship was used after fitting the measured ADC transient to determine tau. The fitted time constant must be in seconds when calculating inductance in henries.

4. Circuit and Hardware Design

The prototype used a small DC motor, ESP32 microcontroller, MOSFET switching stage, Schottky diode, shunt resistor, and op-amp current-sensing circuit. The motor was switched by the ESP32 while the ADC measured the amplified voltage across the shunt resistor.

Component / Parameter	Purpose in Project
3 V DC motor	Device under test; modeled as resistance, inductance, and back-EMF.
ESP32	Generated the switching signal and recorded ADC values at controlled delay points.
MOSFET switch	Controlled motor current from the ESP32 GPIO signal.
Schottky diode	Provided inductive transient protection for the motor switching circuit.
0.127 ohm shunt resistor	Converted motor current into a measurable voltage.
Op-amp sense stage	Amplified the shunt voltage before measurement by the ESP32 ADC.
10 k ohm / 200 k ohm resistor network	Used in the op-amp circuit, giving a nominal gain of approximately 20.
Breadboard wiring	Allowed rapid prototyping and troubleshooting of the low-voltage circuit.



Simplified measurement path: motor current produces a shunt voltage, the op-amp scales it, and the ESP32 samples the ADC response at delayed time

Figure 1. Simplified motor characterization circuit and measurement path.

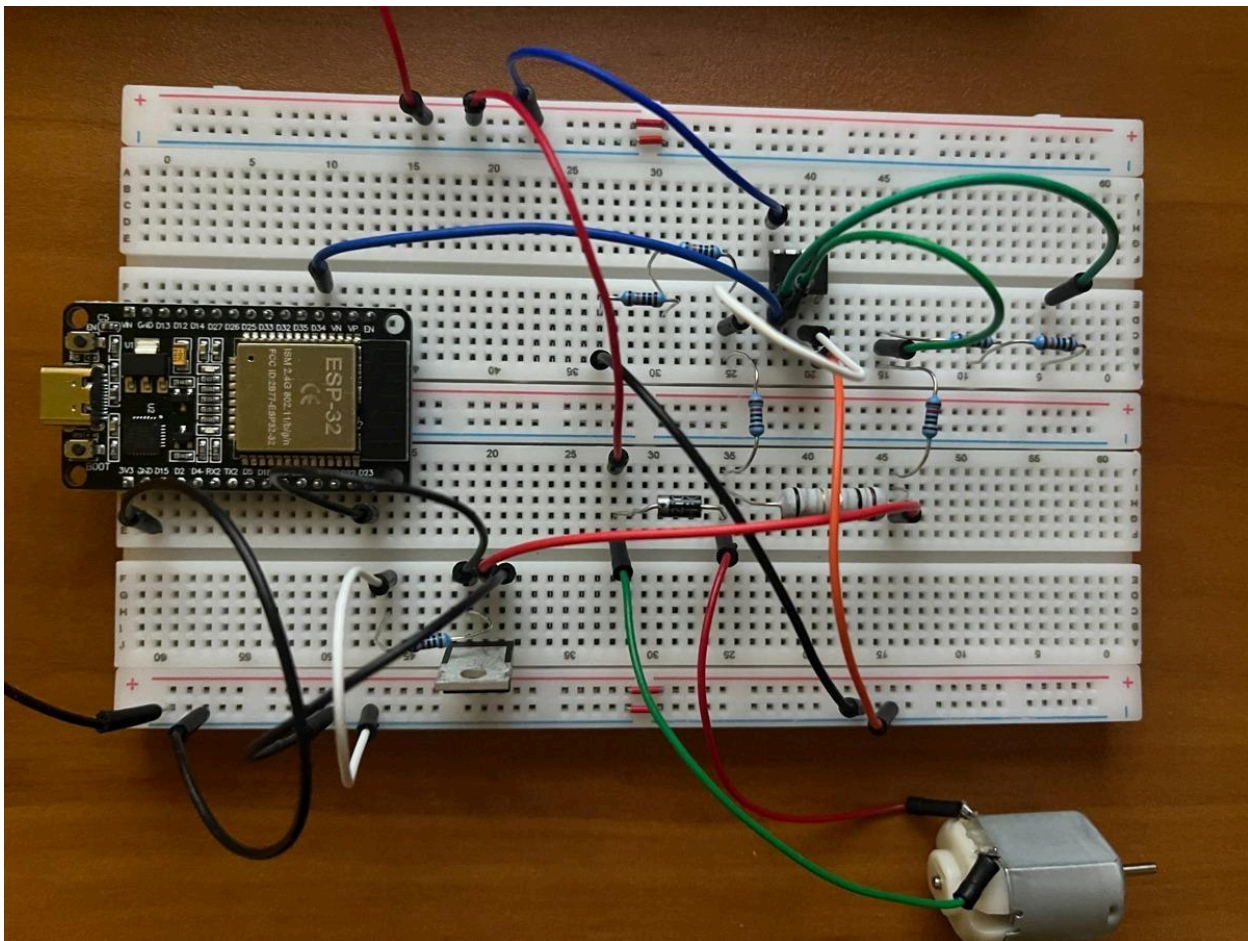


Figure 2. Breadboard prototype used for ESP32-based motor transient measurement.

5. Experimental Method

The test was designed to capture the motor current transient after the motor was switched on. Because the ESP32 sampling rate is limited relative to the short motor time constant, the response was reconstructed by repeating the test at incremental delay values instead of relying on a single continuous capture.

- Set up the motor, switching transistor, shunt resistor, op-amp sensing circuit, and ESP32 ADC input on a breadboard.
- Applied a switching command from the ESP32 GPIO to energize the motor.
- Measured the amplified shunt voltage as an ADC value at selected delay times after turn-on.
- Repeated the test using delay values from 0 us to 250 us in 2 us increments, producing 126 data points.
- Allowed time between samples to reduce heating effects in the motor winding and wiring.
- Imported the collected ADC data into MATLAB for fitting and analysis.

The first point at $t = 0$ us was excluded from the curve fit because it behaved as an initial switching point rather than part of the smooth rising exponential transient. This matches the MATLAB processing script, which fits `adc(2:end)` against `delay_us(2:end)`.

6. Data Processing and Curve Fit

The measured ADC data was fit to a first-order exponential model:

$$y(t) = A(1 - \exp(-t / \tau)) + C$$

In this equation, A is the exponential amplitude in ADC counts, C is the fitted offset, and τ is the time constant. The MATLAB script used `fminsearch` to minimize the sum of squared errors between the measured ADC values and the model.

7. Results

Result / Parameter	Value
Motor test voltage range	3 V
ESP32 / logic supply shown in model	3.3 V
Measured motor/load resistance used for L calculation	3.9 ohm
Shunt resistor	0.127 ohm
Back-EMF value used in simulation model	1.42 V
Initial LTspice/model inductance guess	0.36 mH
Fitted exponential amplitude, A	1010.7 ADC counts
Fitted offset, C	882.9 ADC counts
Fitted final value, $A + C$	1893.6 ADC counts
Fitted time constant, τ	20.3 us
Estimated inductance, $L = R * \tau$	79.2 uH
Fit R-squared	0.939
Fit RMSE	45.0 ADC counts

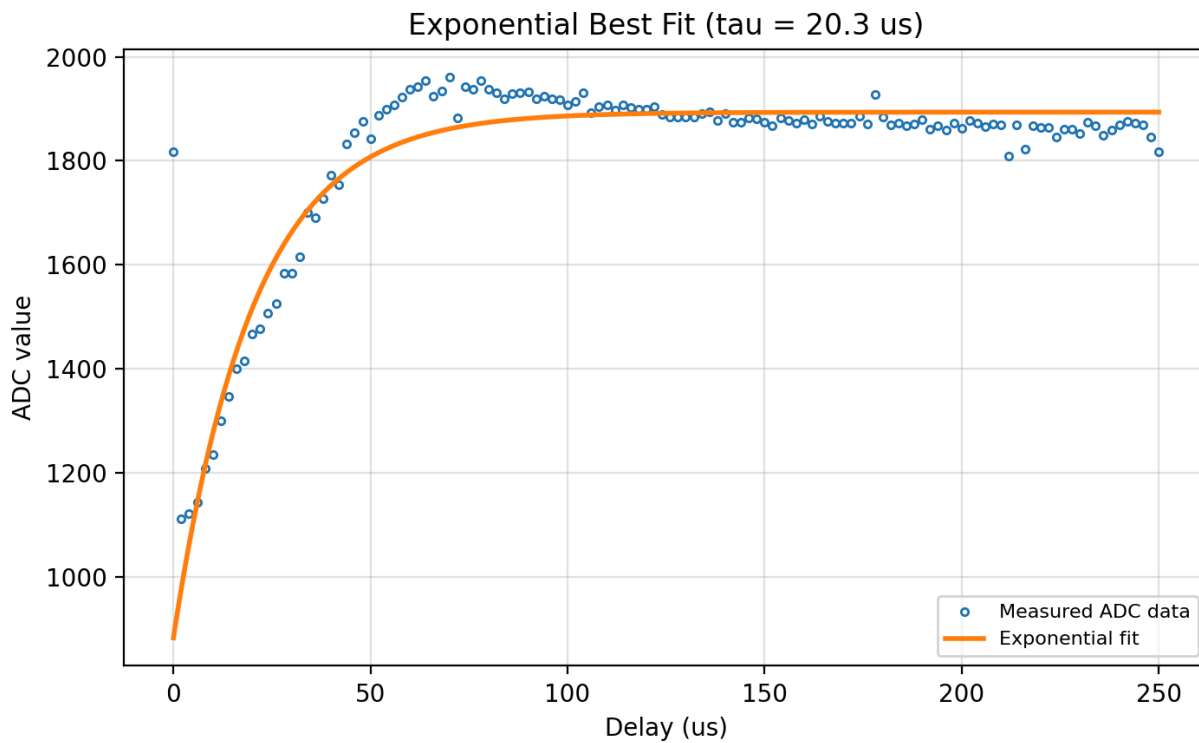


Figure 3. Measured ADC response and exponential best fit. The fitted time constant was 20.3 us.

The fitted time constant was 20.3 us. Converting this to seconds and multiplying by the measured resistance gives:

$$L = R * \tau = 3.9 \text{ ohm} * 20.3\text{e-}6 \text{ s} = 79.2 \text{ uH}$$

The result indicates that the tested motor winding has an estimated inductance of approximately 79 uH under the assumptions of the simplified RL/back-EMF model. The curve fit follows the overall measured transient shape well, but the measured data contains overshoot, noise, and late-time drift that are not captured by the simple first-order model.

8. Discussion

The experimental result supports the use of a first-order transient model for estimating the motor winding inductance. The measured ADC data rises rapidly and approaches a steady value, which is consistent with an RL-type response. The final inductance estimate was lower than the initial 0.36 mH simulation guess, so the measured data should be treated as the stronger value for this specific motor and setup.

The fit quality was acceptable for a breadboard prototype and low-cost ESP32 measurement system. The main source of agreement was the curve shape and time constant. The main sources of mismatch were likely switching artifacts, ADC timing limitations, breadboard parasitics, op-amp bandwidth or offset, motor heating, and simplifying the motor as a fixed back-EMF source during the transient.

9. Challenges and Solutions

Challenge	Effect on Measurement	Solution Used
ESP32 sampling rate	The motor transient occurred on the order of tens of microseconds, which is close to the practical sampling limits of the ESP32 ADC.	Repeated the test at discrete delay intervals and reconstructed the response from multiple runs.
Initial switching artifact	The $t = 0$ us data point did not follow the same exponential trend as the rest of the response.	Excluded the first data point from the fit and used the remaining points for time-constant estimation.
Motor and wire heating	High current during repeated tests can heat the motor winding and change resistance.	Waited between measurements to reduce heating effects before each new sample.
Breadboard parasitics	Loose contacts, long jumper wires, and parasitic inductance/capacitance can affect high-speed measurements.	Kept wiring functional and documented the limitation for future PCB or shorter-wire revisions.
Simplified motor model	A real motor has mechanical acceleration, changing back-EMF, friction, and nonideal switching behavior.	Used the first-order model as an approximation and interpreted the inductance estimate within that assumption.

10. Limitations and Future Improvements

- Calibrate the ADC-to-current conversion more rigorously using known currents and a verified op-amp gain measurement.
- Use an oscilloscope or higher-speed data-acquisition system to capture the transient directly in a single run.
- Measure the op-amp offset, bandwidth, and saturation behavior to improve confidence in the ADC signal.
- Replace the breadboard with a compact PCB or soldered prototype to reduce parasitics and wiring variation.
- Repeat tests at multiple supply voltages and loads to compare how the estimated inductance changes with operating condition.
- Model the motor mechanically by including speed-dependent back-EMF rather than holding back-EMF constant.

11. Conclusion

This project successfully demonstrated a practical method for estimating the inductance of a small DC motor using a low-voltage test circuit, ESP32 sampling, and exponential curve fitting. The fitted transient response produced a time constant of 20.3 us. With a measured resistance of 3.9 ohm, the estimated inductance was 79.2 uH.

The results were consistent in overall curve shape with the expected first-order behavior. Although the setup had limitations due to sampling rate, heating, breadboard wiring, and model simplifications, the project produced a repeatable measurement approach and clear documentation for future improvement.

Appendix A - MATLAB Fitting Script

The following script was used to store the measured ADC data, fit the response to an exponential curve, extract the time constant, and generate the plot.

```
delay_us = 0:2:250;

% ADC data
adc = [ ...
    1817,1111,1122,1143,1209,1235,1301,1347,1401,1415, ...
    1467,1477,1507,1526,1584,1584,1616,1701,1691,1728, ...
    1773,1754,1833,1855,1876,1843,1888,1900,1908,1923, ...
    1938,1943,1955,1925,1935,1961,1883,1943,1937,1955, ...
    1937,1931,1920,1929,1931,1932,1920,1925,1920,1917, ...
    1907,1915,1931,1892,1904,1907,1897,1907,1903,1899, ...
    1899,1904,1890,1885,1885,1885,1884,1891,1894,1878, ...
    1891,1875,1874,1882,1881,1874,1867,1883,1877,1873, ...
    1879,1871,1886,1876,1872,1872,1872,1886,1871,1927, ...
```

```

1884,1869,1872,1868,1871,1879,1861,1867,1860,1873, ...
1862,1878,1872,1866,1871,1869,1809,1869,1823,1867, ...
1865,1865,1846,1861,1861,1853,1875,1867,1850,1860, ...
1869,1876,1872,1869,1846,1818 ];

t = delay_us(2:end);
y = adc(2:end);

model = @(b,t) b(1) * (1 - exp(-t / b(2))) + b(3);

% Find Fits
cost = @(b) sum((model(b,t) - y).^2);

% Initial guess
b0 = [800, 80, 1100];

% Minimize
b = fminsearch(cost, b0);

A = b(1);
tau = b(2);
C = b(3);

% Create Fit Curve
t_fit = linspace(0, 250, 500);
y_fit = model(b, t_fit);

% Generate it
figure;
plot(delay_us, adc, 'o', 'MarkerSize', 4); hold on;
plot(t_fit, y_fit, 'LineWidth', 2);
grid on;

xlabel('Delay (\mus)');
ylabel('ADC Value');
title(sprintf('Exponential Best Fit (\tau = %.1f \mus)', tau));
legend('Measured Data', 'Exponential Fit', 'Location', 'best');

```

Appendix B - Raw ADC Data

The raw data below records the delay after motor turn-on and the corresponding ESP32 ADC value. The data was collected from 0 us to 250 us in 2 us increments.

Delay (us)	ADC	Delay (us)	ADC	Delay (us)	ADC
0	1817	2	1111	4	1122
6	1143	8	1209	10	1235
12	1301	14	1347	16	1401
18	1415	20	1467	22	1477
24	1507	26	1526	28	1584
30	1584	32	1616	34	1701
36	1691	38	1728	40	1773
42	1754	44	1833	46	1855
48	1876	50	1843	52	1888
54	1900	56	1908	58	1923
60	1938	62	1943	64	1955
66	1925	68	1935	70	1961
72	1883	74	1943	76	1937
78	1955	80	1937	82	1931
84	1920	86	1929	88	1931
90	1932	92	1920	94	1925
96	1920	98	1917	100	1907
102	1915	104	1931	106	1892
108	1904	110	1907	112	1897
114	1907	116	1903	118	1899
120	1899	122	1904	124	1890
126	1885	128	1885	130	1885

132	1884	134	1891	136	1894
138	1878	140	1891	142	1875
144	1874	146	1882	148	1881
150	1874	152	1867	154	1883
156	1877	158	1873	160	1879
162	1871	164	1886	166	1876
168	1872	170	1872	172	1872
174	1886	176	1871	178	1927
180	1884	182	1869	184	1872
186	1868	188	1871	190	1879
192	1861	194	1867	196	1860
198	1873	200	1862	202	1878
204	1872	206	1866	208	1871
210	1869	212	1809	214	1869
216	1823	218	1867	220	1865
222	1865	224	1846	226	1861
228	1861	230	1853	232	1875
234	1867	236	1850	238	1860
240	1869	242	1876	244	1872
246	1869	248	1846	250	1818